



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	NUCLEAR AND PARTICLE PHYSICS				
Type of Course	Core in Major				
Semester	VI				
Academic Level	300 - 399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	Strong foundation in classical mechanics, electromagnetism, quantum mechanics, and mathematics along with a basic understanding of modern physics concepts.				
Course Summary	The course in nuclear and particle physics provides an in-depth exploration of the fundamental constituents of matter, their interactions, and the underlying principles governing nuclear structure, particle behavior, and their implications in theoretical and experimental physics.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the fundamental principles of nuclear and particle physics.	Understand	Conceptual Knowledge	Quizzes, Tests

CO2	Analyse nuclear structure and properties, including nuclear forces and decay processes.	Analyse	Procedural Knowledge	Homework Assignments
CO3	Apply theoretical models to predict nuclear reactions and particle behavior.	Apply	Conceptual Knowledge	Problem Sets, Projects
CO4	Analyse the processes and mechanisms of radioactive decay.	Analyse	Procedural Knowledge	Homework, Exams
CO5	Describe the operation and components of particle accelerators.	Understand	Basic Concepts	Virtual lab Demonstrations
CO6	Analyse the principles and techniques of particle Detectors.	Analyse	Conceptual Knowledge	Problem Sets, Exams
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48 +12)	Marks (70)
I	NUCLEAR PROPERTIES AND NUCLEAR MODELS		15	20
	1	Introduction	1	
	2	Quantitative Facts About Nucleus	2	
	3	Binding Energy	2	
	4	Nuclear Angular Momentum; Nuclear Moments; Parity	3	
	5	Nuclear Force	1	
	6	Liquid Drop model	3	
	7	Shell Model	3	
	Sections 1.1 – 1.10 of chapter 1; sections 2.1 – 2.3 of chapter 2 of Book 1			

II	RADIOACTIVITY		15	20
	8	Introduction	1	
	9	Laws of Disintegration; Radioactive Series	3	
	10	Alpha Emission	3	
	11	Beta Decay	4	
	12	Gamma Decay	2	
	13	Artificial or Induced Radioactivity; Applications of Radioactivity	2	
	Sections 3.1, 3.2, 3.5 – 3.8 of Book 1			
III	NUCLEAR REACTIONS		9	15
	14	Nuclear Reaction Cross-section	1	
	15	Conservation Laws in Nuclear Reactions; Kinematics of Nuclear Reactions; Compound Nucleus	3	
	16	Nuclear Fission	2	
	17	Nuclear Fusion	2	
	18	Interaction of Gamma Rays with Matter	1	
	Sections 4.1, 4.3 – 4.8, 5.5 of Book 1			
IV	PARTICLE PHYSICS		9	15
	19	Types of Interactions	1	
	20	Classification of Elementary Particles	1	
	21	Quantum Numbers	2	
	22	Conservation Laws; Weak Decays of Strange Particles	3	
	23	Quarks; Qualitative Description of Quark Model	2	
	Sections 8.3, 8.4, 8.6 – 8.8, 8.10, 8.11 of chapter 8 of Book 1			
V	OPEN-ENDED MODULE: PARTICLE ACCELERATORS AND RADIATION DETECTORS		12	
Books and References:				

1. Introduction to Nuclear and Particle Physics - V K Mittal, R C Verma and S C Gupta (Book 1)
2. Introductory Nuclear Physics by Kenneth S. Krane
3. Concepts of Modern Physics by Arthur Beiser
4. Nuclear and Particle Physics: An Introduction by Brian R. Martin and Graham Shaw
5. Nuclear and Particle Physics: An Introduction by S. N. Ghoshal and T. K. Basak
6. Nuclear Physics: Theory and Experiment by Raj Kumar Gupta

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO 3	PO 4	PO 5	P O6	PO 7
CO 1	2	2	3	2	0	2	2	1	2	2	0	2	2
CO 2	3	2	2	3	2	0	3	0	0	2	2	0	0
CO 3	3	3	3	2	0	2	2	2	1	0	2	1	2
CO 4	2	2	2	2	0	0	3	0	2	2	0	0	0
CO 5	2	3	2	2	1	3	1	2	0	2	1	2	2
CO 6	0	2	2	2	0	2	0	0	2	0	0	2	0

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory / Practical Exam
- Assignments / Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory / Practical Exam	Assignment / Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	